Vaughn Khouri

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Education

Massachusetts Institute of Technology (MIT) Cambridge, MA
B.S. Mechanical Engineering, Concentration in Control, Instrumentation & Robotics (2A-6) *Expected Jun. 2028*

- Planned Minor in Materials Science and Engineering (Course 3)
- GPA: 5.0/5.0

Experience

Brakes Subsystem, Dynamometer Lead Sept. 2025 – Present
MIT Motorsports FSAE Team *Cambridge, MA*

- Lead the design (SolidWorks) and fabrication (CNC lathe, mill) of a dynamometer platform to characterize motors and custom inverters under prolonged load. Sized for accurate torque measurements at a peak output power of 40 kW and 350 Hz driving frequency, verified via FEA.
- Design and manufacture custom brake rotors and mounting buttons (Siemens NX, CNC lathe, waterjet).
  - Reduced weight and size of MY26 brake mounting buttons by 50% through improved sizing calculations, enabling smaller carbon fiber rims and reducing overall vehicle mass by 4 kg.
- Write and lead design reviews with drivetrain team members and alumni.

Founding Engineer, Platform Development May 2025 – Sept. 2025
Acre Robotics *Redwood City, CA*

- Led hardware development from Seed stage to Series A, working closely with CV and Controls teams from day one.
- Owned design, installation, and bring-up of custom hardware (SolidWorks) and electronics (Jetson, ESP32, Stereolabs Zed, Ouster LiDAR), enabling the first autonomous driving tests on a full-size Case 75A tractor.
  - Installed and fine-tuned RTK GPS hardware, achieving real-time accuracy within 5 cm.
  - Reverse-engineered the tractor ECU and other onboard auxiliary devices to interface with 10+ OEM sensors.
- Completed and shipped a full-featured MVP for stress testing on an operational farm in under 30 days.

Mechatronics Intern Summer 2023, 2024
Apple (ETC, AEC II) *Cupertino, CA*

- Selected as one of 15 students for two summer programs at Apple. Led 5-person teams to design (Onshape) and fabricate functional prototypes of novel consumer products, create presentations, and execute live demos during stage pitches to John Ternus and other Apple executives. Ex:
  - "Flex," a portable cable workout machine that sticks to surfaces. Designed (Onshape) a compact active suction system with a BLDC motor, enabling exercises with up to 40 lbs of resistance. Integrated custom electronics (Pi Pico, ODrive) and developed firmware to interface with a companion app via Bluetooth.
- Collaborated with Apple engineers and project managers.

Team Captain Sept. 2021 – Jul. 2025
FIRST Robotics Competition Team 8033 *Piedmont, CA*

- Drove growth from founding season to a robust organization with 50+ students from 17+ high schools, placing in the top 16 worldwide (2024) and top 15 in California (2022–2025).
- Prototyped, designed, and iterated robot hardware (Onshape); set design requirements; led training and all-team meetings; created and documented design standards and workflows; and drove robot at 4 world championships.
- Designed (Onshape) and coordinated fabrication of a cost-effective, full-size (54' × 26') testing arena constructed from modular welded steel box tube sections — one of only three in California.

Technical Skills

CAD / CAM Tools: PTC Onshape, Siemens NX, SolidWorks, Fusion 360, Inventor, CATIA (including GD&T)
Machine Tools: Manual/CNC Mill, Lathe, MIG & TIG Welding, Waterjet / Fiber Laser, FDM / SLA / SLS 3D Printing, All Common Wood / Metalworking Tools
Other: Conversational Mandarin (reading & writing), First Aid, outdoormanship (BSA Eagle Scout), Microsoldering & Basic PCB Design, Custom Battery Pack Design / Fabrication

More info and full project writeups available on my portfolio: https://www.vkhouri.dev